# Multivariable Calculus: Problem Set 8

Detailed Assignment-Style Solutions with Diagrams

Prepared solution set

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## Conventions used throughout

The goal of these solutions is to show the full calculation path, not only the final answer. For every optimization problem, I first identify the feasible set, reduce the number of variables whenever possible, and then check whether maxima/minima exist from compactness or from an unbounded direction. For every integral problem, I first write the region or solid explicitly, because a wrong region almost always gives a wrong integral.

**Method / reasoning plan**

For constrained extrema with one or two constraints, I use one of two methods.

1. If the constraints allow direct substitution, I reduce to fewer variables and optimize the reduced expression.
2. If direct substitution is less convenient, I use Lagrange multipliers: at an interior constrained extremum, the gradient of the objective lies in the span of the gradients of the active constraints.

For integrals, I always write the region in inequalities before integrating. This is the main check that prevents accidental reversal of limits.

# 1 Problem 1: Extreme values with two constraints

## 1.1 Problem 1(i)

Find the extreme values of

$$f(x, y, z) = x + 2y$$

subject to

$$x + y + z = 1, \qquad y^2 + z^2 = 4.$$

**Method / reasoning plan**

The first constraint is a plane and the second constraint is a circular cylinder in the $yz$-variables. The plane lets us solve for $x$. Then the objective becomes a linear function on the circle $y^2 + z^2 = 4$. A linear function on a circle is maximized in the direction of its coefficient vector and minimized in the opposite direction.

From the plane,

$$x = 1 - y - z.$$

Substitute this into $f$:

$$f = x + 2y = (1 - y - z) + 2y = 1 + y - z.$$

Thus we need to maximize and minimize

$$1 + y - z$$

subject to

$$y^2 + z^2 = 4.$$

The variable part is

$$y - z = (1, -1) \cdot (y, z).$$

The vector $(1, -1)$ has length

$$\sqrt{1^2 + (-1)^2} = \sqrt{2}.$$

On the circle of radius 2, the largest possible value of $(1, -1) \cdot (y, z)$ is

$$2\sqrt{2},$$

and the smallest possible value is

$$-2\sqrt{2}.$$

Therefore,

$$f_{\max} = 1 + 2\sqrt{2}, \qquad f_{\min} = 1 - 2\sqrt{2}.$$

The maximum occurs when $(y, z)$ points in the direction $(1, -1)$:

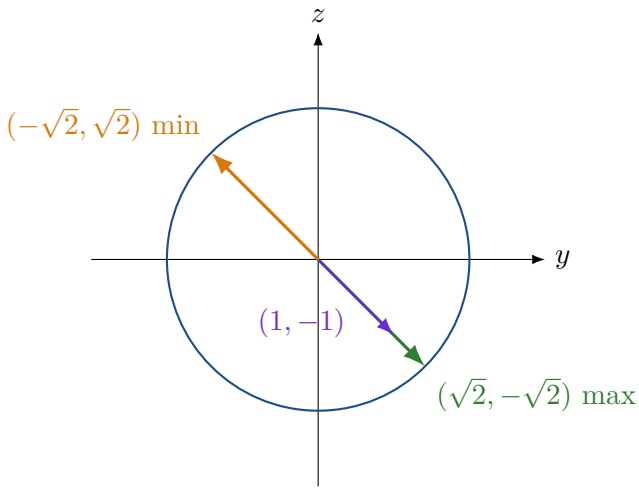
$$(y, z) = 2\frac{(1, -1)}{\sqrt{2}} = (\sqrt{2}, -\sqrt{2}).$$

Then

$$x = 1 - y - z = 1 - \sqrt{2} + \sqrt{2} = 1.$$

The minimum occurs at the opposite point:

$$(y, z) = (-\sqrt{2}, \sqrt{2}), \qquad x = 1.$$




Circle $y^2 + z^2 = 4$ and objective direction $y - z$.

Figure 1: Reduction of Problem 1(i) to a linear objective on a circle in the $(y, z)$-plane.

**Final answer**

$$\boxed{f_{\max} = 1 + 2\sqrt{2} \text{ at } (1, \sqrt{2}, -\sqrt{2})}$$

$$\boxed{f_{\min} = 1 - 2\sqrt{2} \text{ at } (1, -\sqrt{2}, \sqrt{2})}$$

## 1.2 Problem 1(ii)

Find the extreme values of

$$f(x, y, z) = 3x - y - 3z$$

subject to

$$x + y - z = 0, \qquad x^2 + 2z^2 = 1.$$

The plane gives

$$y = z - x.$$

Substitute into $f$:

$$f = 3x - (z - x) - 3z = 4x - 4z = 4(x - z).$$

Thus the problem reduces to maximizing/minimizing

$$4x - 4z$$

subject to

$$x^2 + 2z^2 = 1.$$

Write this as an ellipse. We optimize the linear form

$$a \cdot (x, z), \qquad a = (4, -4),$$

under the quadratic constraint

$$(x, z)^T Q(x, z) = 1, \qquad Q = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

For an ellipse of the form $u^T Q u = 1$, the maximum value of $a \cdot u$ is

$$\sqrt{a^T Q^{-1} a}.$$

Here

$$Q^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1/2 \end{pmatrix},$$

so

$$a^T Q^{-1} a = 4^2 + (-4)^2 \frac{1}{2} = 16 + 8 = 24.$$

Therefore

$$f_{\max} = \sqrt{24} = 2\sqrt{6}, \qquad f_{\min} = -2\sqrt{6}.$$

The maximizing point in $(x, z)$ is

$$(x, z) = \frac{Q^{-1} a}{\sqrt{a^T Q^{-1} a}} = \frac{(4, -2)}{2\sqrt{6}} = \left( \frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}} \right).$$

Then

$$y = z - x = -\frac{1}{\sqrt{6}} - \frac{2}{\sqrt{6}} = -\frac{3}{\sqrt{6}} = -\frac{\sqrt{6}}{2}.$$

The minimizing point is the negative of this point in the $(x, z)$-ellipse:

$$(x, z) = \left( -\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right), \qquad y = \frac{\sqrt{6}}{2}.$$

**Final answer**

$$\boxed{f_{\max} = 2\sqrt{6} \text{ at } \left( \frac{2}{\sqrt{6}}, -\frac{\sqrt{6}}{2}, -\frac{1}{\sqrt{6}} \right)}$$

$$\boxed{f_{\min} = -2\sqrt{6} \text{ at } \left( -\frac{2}{\sqrt{6}}, \frac{\sqrt{6}}{2}, \frac{1}{\sqrt{6}} \right)}$$

### 1.3 Problem 1(iii)

Find the extreme values of

$$f(x, y, z) = yz + xy$$

subject to

$$xy = 1, \qquad y^2 + z^2 = 1.$$

The constraint $xy = 1$ gives

$$x = \frac{1}{y},$$

so $y \neq 0$. Then

$$f = yz + xy = yz + 1.$$

Thus the problem is equivalent to maximizing/minimizing $yz + 1$ on the circle

$$y^2 + z^2 = 1.$$

For real numbers $y, z$,

$$(y - z)^2 \geq 0$$

implies

$$y^2 + z^2 - 2yz \geq 0.$$

Since $y^2 + z^2 = 1$, we get

$$yz \leq \frac{1}{2}.$$

Similarly,

$$(y + z)^2 \geq 0$$

implies

$$y^2 + z^2 + 2yz \geq 0,$$

so

$$yz \geq -\frac{1}{2}.$$

Therefore

$$\frac{1}{2} \leq yz + 1 \leq \frac{3}{2}.$$

The maximum $yz = 1/2$ occurs when $y = z = 1/\sqrt{2}$ or $y = z = -1/\sqrt{2}$. The minimum $yz = -1/2$ occurs when $z = -y$ and $|y| = 1/\sqrt{2}$.

**Final answer**

$$\boxed{f_{\max} = \frac{3}{2}}$$

attained at

$$\boxed{(x, y, z) = \left(\sqrt{2}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \left(-\sqrt{2}, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)}.$$

$$\boxed{f_{\min} = \frac{1}{2}}$$

attained at

$$\boxed{(x, y, z) = \left(\sqrt{2}, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \left(-\sqrt{2}, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)}.$$

### 1.4 Problem 1(iv)

Find the extreme values of

$$f(x, y, z) = x^2 + y^2 + z^2$$

subject to

$$x - y = 1, \qquad y^2 - z^2 = 1.$$

The first constraint gives

$$x = y + 1.$$

The second gives

$$z^2 = y^2 - 1.$$

Therefore feasible points must satisfy $|y| \geq 1$. Substitute into $f$:

$$f = (y + 1)^2 + y^2 + z^2.$$

Since $z^2 = y^2 - 1$,

$$f = (y + 1)^2 + y^2 + (y^2 - 1) = y^2 + 2y + 1 + y^2 + y^2 - 1 = 3y^2 + 2y.$$

Thus we minimize

$$F(y) = 3y^2 + 2y$$

on the noncompact set

$$(-\infty, -1] \cup [1, \infty).$$

This quadratic opens upward. Its unconstrained vertex is at

$$F'(y) = 6y + 2 = 0 \quad \Longrightarrow \quad y = -\frac{1}{3},$$

but $-1/3$ is not feasible because $|y| \geq 1$. Therefore the minimum occurs at the feasible endpoint closest to $-1/3$, namely $y = -1$. Then

$$z^2 = (-1)^2 - 1 = 0, \qquad z = 0, \qquad x = y + 1 = 0.$$

The minimum value is

$$f(0, -1, 0) = 0^2 + (-1)^2 + 0^2 = 1.$$

There is no maximum because as $|y| \to \infty$,

$$f = 3y^2 + 2y \to \infty.$$

**Final answer**

$$\boxed{f_{\min} = 1 \text{ at } (0, -1, 0).}$$

$$\boxed{\text{There is no maximum value because the feasible set is unbounded and } f \to \infty.}$$

## 2 Problem 2: Global maxima and minima on closed regions

### 2.1 Problem 2(i)

Find the absolute maximum and minimum of

$$f(x, y) = 4xy^2 - x^2y^2 - xy^3$$

on the closed triangular region with vertices $(0, 0), (0, 6), (6, 0)$.

**Method / reasoning plan**

Because the region is closed and bounded and $f$ is continuous, absolute maxima and minima exist. We must check interior critical points and all boundary pieces.

Factor the function:

$$f(x,y) = xy^2(4-x-y).$$

The triangular region is

$$R = \{(x,y) : x \geq 0,\ y \geq 0,\ x+y \leq 6\}.$$

Interior critical points satisfy $f_x = 0$ and $f_y = 0$. Compute:

$$f_x = y^2(4-2x-y),$$

and

$$f_y = x\left(2y(4-x-y) - y^2\right) = xy(8-2x-3y).$$

In the interior, $x > 0$ and $y > 0$, so

$$4-2x-y = 0, \qquad 8-2x-3y = 0.$$

From the first equation,

$$y = 4-2x.$$

Substitute into the second equation:

$$8-2x-3(4-2x) = 0.$$

So

$$8-2x-12+6x = 0 \quad\Longrightarrow\quad 4x-4 = 0 \quad\Longrightarrow\quad x = 1.$$

Then

$$y = 4-2(1) = 2.$$

The interior critical value is

$$f(1,2) = 1\cdot 2^2(4-1-2) = 4.$$

Now check the boundary.

**Boundary 1:** $x = 0$**.** Then

$$f(0,y) = 0.$$

**Boundary 2:** $y = 0$**.** Then

$$f(x,0) = 0.$$

**Boundary 3:** $x+y = 6$**.** Put $y = 6-x$, where $0 \leq x \leq 6$. Then

$$f = x(6-x)^2(4-6) = -2x(6-x)^2.$$

Let

$$h(x) = -2x(6-x)^2.$$

Differentiate:

$$h'(x) = -2\left[(6-x)^2 + x\cdot 2(6-x)(-1)\right].$$

Factor:

$$h'(x) = -2(6-x)\left[(6-x)-2x\right] = -2(6-x)(6-3x).$$

Thus the boundary critical points are

$$x = 6 \quad \text{or} \quad x = 2.$$

At $x = 6$, $y = 0$, and $f = 0$. At $x = 2$, $y = 4$, and

$$f(2,4) = 2 \cdot 16(4 - 2 - 4) = 32(-2) = -64.$$

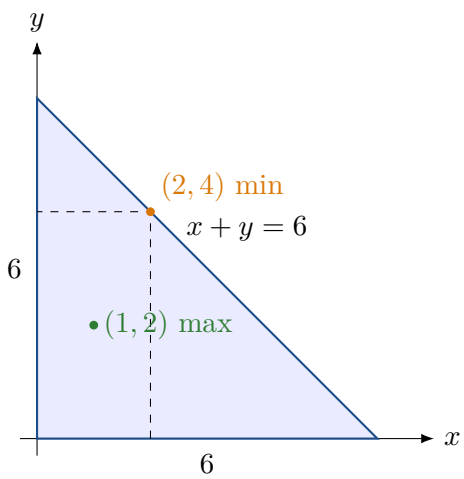



Figure 2: Triangular region for Problem 2(i). The interior point gives the maximum and a boundary point gives the minimum.

**Final answer**

$$\boxed{\text{Absolute maximum } = 4 \text{ at } (1,2).}$$

$$\boxed{\text{Absolute minimum } = -64 \text{ at } (2,4).}$$

## 2.2 Problem 2(ii)

Find the absolute maximum and minimum of

$$f(x,y) = e^{-x^2-y^2}(x^2 + 2y^2)$$

on

$$R = \{(x,y) : x^2 + y^2 \leq 1\}.$$

Use polar coordinates:

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad 0 \leq r \leq 1.$$

Then

$$x^2 + y^2 = r^2,$$

and

$$x^2 + 2y^2 = r^2\cos^2\theta + 2r^2\sin^2\theta = r^2(\cos^2\theta + 2\sin^2\theta).$$

Since $\cos^2\theta = 1 - \sin^2\theta$,

$$x^2 + 2y^2 = r^2(1 + \sin^2\theta).$$

Therefore

$$f = e^{-r^2}r^2(1 + \sin^2\theta).$$

For fixed $r$, the factor $1 + \sin^2\theta$ ranges from 1 to 2. The function is always nonnegative, so the minimum is clearly

$$f(0,0) = 0.$$

For the maximum, maximize

$$2r^2e^{-r^2}$$

on $0 \le r \le 1$. Let $s = r^2$. Then $0 \le s \le 1$, and

$$se^{-s}$$

has derivative

$$e^{-s} - se^{-s} = e^{-s}(1-s).$$

This is nonnegative on $[0,1]$ and zero only at $s = 1$. Therefore the maximum occurs when

$$r = 1$$

and

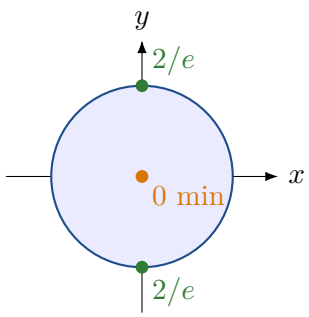
$$\sin^2\theta = 1.$$

Thus

$$(x,y) = (0,1) \quad \text{or} \quad (0,-1).$$

The maximum value is

$$2e^{-1} = \frac{2}{e}.$$




Unit disk; max lies on the $y$-axis where $\sin^2\theta = 1$.

Figure 3: Disk region for Problem 2(ii).

**Final answer**

$$\boxed{\text{Absolute minimum } = 0 \text{ at } (0,0).}$$

$$\boxed{\text{Absolute maximum } = \frac{2}{e} \text{ at } (0,1) \text{ and } (0,-1).}$$

# 3 Problem 3: Double integrals as volumes

## 3.1 Problem 3(i)

Evaluate

$$\iint_R 3\,dA,$$

where

$$R = \{(x,y) : -2 \le x \le 2,\ 1 \le y \le 6\}.$$

**Source/interpretation note**

The source scan is visually consistent with $-2 \le x \le 2$. If the lower bound were literally $2 \le x \le 2$, the region would have zero width. I use the mathematically intended rectangular region $-2 \le x \le 2$.

The integrand 3 is a constant height above the rectangle. The area of $R$ is

$$(2-(-2))(6-1) = 4 \cdot 5 = 20.$$

Therefore the volume is

$$3 \cdot 20 = 60.$$

**Final answer**

$$\boxed{\iint_R 3\,dA = 60.}$$

### 3.2 Problem 3(ii)

Evaluate

$$\iint_R 5x\,dA,$$

where

$$R = \{(x,y) : 0 \le x \le 5,\ 0 \le y \le 3\}.$$

Write the integral explicitly:

$$\iint_R 5x\,dA = \int_0^5 \int_0^3 5x\,dy\,dx.$$

First integrate with respect to $y$:

$$\int_0^3 5x\,dy = 5x[y]_0^3 = 15x.$$

Then integrate with respect to $x$:

$$\int_0^5 15x\,dx = 15\left[\frac{x^2}{2}\right]_0^5 = 15 \cdot \frac{25}{2} = \frac{375}{2}.$$

**Final answer**

$$\boxed{\iint_R 5x\,dA = \frac{375}{2}.}$$

## 4 Problem 4: Iterated integrals

**Method / reasoning plan**

For each iterated integral, I integrate from the inside outward. Constants with respect to the current variable are pulled outside the inner integral, and every antiderivative is evaluated at both bounds.

### 4.1 Problem 4(i)

$$I = \int_0^2 \int_0^4 y^2 e^{2x}\,dy\,dx.$$

Since $e^{2x}$ is constant with respect to $y$,

$$\int_0^4 y^2 e^{2x}\,dy = e^{2x}\left[\frac{y^3}{3}\right]_0^4 = e^{2x}\frac{64}{3}.$$

Therefore

$$I = \frac{64}{3}\int_0^2 e^{2x}\,dx = \frac{64}{3}\left[\frac{1}{2}e^{2x}\right]_0^2 = \frac{32}{3}(e^4-1).$$

$$\boxed{I = \frac{32}{3}(e^4-1).}$$

## 4.2 Problem 4(ii)

$$I = \int_{-3}^{3}\int_0^{\pi/2}(y + y^2\cos x)\,dx\,dy.$$

Inner integral:

$$\int_0^{\pi/2}(y + y^2\cos x)\,dx = y\,[x]_0^{\pi/2} + y^2\,[\sin x]_0^{\pi/2} = \frac{\pi y}{2} + y^2.$$

Thus

$$I = \int_{-3}^{3}\left(\frac{\pi y}{2} + y^2\right)dy.$$

The term $\frac{\pi y}{2}$ is odd on $[-3,3]$, so its integral is zero. Hence

$$I = \int_{-3}^{3} y^2\,dy = \left[\frac{y^3}{3}\right]_{-3}^{3} = 9-(-9) = 18.$$

$$\boxed{I = 18.}$$

## 4.3 Problem 4(iii)

$$I = \int_1^3\int_1^5 xy\ln y\,dy\,dx.$$

Since $x$ is constant in the inner integral,

$$I = \int_1^3 x\left(\int_1^5 y\ln y\,dy\right)dx.$$

Use integration by parts for $\int y\ln y\,dy$. Let

$$u = \ln y, \qquad dv = y\,dy.$$

Then

$$du = \frac{1}{y}\,dy, \qquad v = \frac{y^2}{2}.$$

So

$$\int y\ln y\,dy = \frac{y^2}{2}\ln y - \int\frac{y^2}{2}\cdot\frac{1}{y}\,dy = \frac{y^2}{2}\ln y - \int\frac{y}{2}\,dy = \frac{y^2}{2}\ln y - \frac{y^2}{4}.$$

Evaluate from 1 to 5:

$$\int_1^5 y\ln y\,dy = \left[\frac{y^2}{2}\ln y - \frac{y^2}{4}\right]_1^5 = \frac{25}{2}\ln 5 - \frac{25}{4} + \frac{1}{4} = \frac{25}{2}\ln 5 - 6.$$

Therefore

$$I = \left(\frac{25}{2}\ln 5 - 6\right)\int_1^3 x\,dx = \left(\frac{25}{2}\ln 5 - 6\right)\left[\frac{x^2}{2}\right]_1^3.$$

Since

$$\left[\frac{x^2}{2}\right]_1^3 = \frac{9-1}{2} = 4,$$

we get

$$I = 50\ln 5 - 24.$$

$$\boxed{I = 50\ln 5 - 24.}$$

### 4.4 Problem 4(iv)

$$I = \int_0^1 \int_0^1 v(u+v^2)^4 \, du \, dv.$$

For fixed $v$, use $w = u + v^2$, so $dw = du$. Then

$$\int_0^1 v(u+v^2)^4 \, du = v \left[ \frac{(u+v^2)^5}{5} \right]_0^1 = \frac{v}{5} \left( (1+v^2)^5 - v^{10} \right).$$

Thus

$$I = \frac{1}{5} \int_0^1 v(1+v^2)^5 \, dv - \frac{1}{5} \int_0^1 v^{11} \, dv.$$

For the first integral, let $s = 1 + v^2$, so $ds = 2v \, dv$. When $v = 0$, $s = 1$. When $v = 1$, $s = 2$. Hence

$$\int_0^1 v(1+v^2)^5 \, dv = \frac{1}{2} \int_1^2 s^5 \, ds = \frac{1}{2} \left[ \frac{s^6}{6} \right]_1^2 = \frac{64-1}{12} = \frac{21}{4}.$$

The second integral is

$$\int_0^1 v^{11} \, dv = \frac{1}{12}.$$

Therefore

$$I = \frac{1}{5} \left( \frac{21}{4} - \frac{1}{12} \right) = \frac{1}{5} \left( \frac{63-1}{12} \right) = \frac{1}{5} \cdot \frac{62}{12} = \frac{31}{30}.$$

$$\boxed{I = \frac{31}{30}.}$$

### 4.5 Problem 4(v)

$$I = \int_0^1 \int_0^1 xy\sqrt{x^2+y^2} \, dy \, dx.$$

For fixed $x$, compute the inner integral:

$$\int xy\sqrt{x^2+y^2} \, dy = x \int y\sqrt{x^2+y^2} \, dy.$$

Let

$$s = x^2 + y^2, \qquad ds = 2y \, dy.$$

Then

$$x \int y\sqrt{x^2+y^2} \, dy = \frac{x}{2} \int s^{1/2} \, ds = \frac{x}{2} \cdot \frac{2}{3} s^{3/2} = \frac{x}{3}(x^2+y^2)^{3/2}.$$

Therefore

$$\int_0^1 xy\sqrt{x^2+y^2} \, dy = \frac{x}{3} \left( (x^2+1)^{3/2} - x^3 \right).$$

Now integrate in $x$:

$$I = \frac{1}{3} \int_0^1 x(x^2+1)^{3/2} \, dx - \frac{1}{3} \int_0^1 x^4 \, dx.$$

For the first integral, let $s = x^2 + 1$, so $ds = 2x \, dx$. Then

$$\int_0^1 x(x^2+1)^{3/2} \, dx = \frac{1}{2} \int_1^2 s^{3/2} \, ds = \frac{1}{2} \left[ \frac{2}{5} s^{5/2} \right]_1^2 = \frac{1}{5}(2^{5/2} - 1).$$

Also,

$$\int_0^1 x^4 \, dx = \frac{1}{5}.$$

Thus

$$I = \frac{1}{3} \left( \frac{2^{5/2}-1}{5} - \frac{1}{5} \right) = \frac{2^{5/2}-2}{15}.$$

$$\boxed{I = \frac{2^{5/2}-2}{15}.}$$

### 4.6 Problem 4(vi)

$$I = \int_0^2 \int_0^\pi r \sin^2\theta \, d\theta \, dr.$$

Separate the variables:

$$I = \left(\int_0^2 r \, dr\right)\left(\int_0^\pi \sin^2\theta \, d\theta\right).$$

The first integral is

$$\int_0^2 r \, dr = \left[\frac{r^2}{2}\right]_0^2 = 2.$$

The standard identity

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$

gives

$$\int_0^\pi \sin^2\theta \, d\theta = \int_0^\pi \frac{1-\cos 2\theta}{2} \, d\theta = \frac{\pi}{2}.$$

Therefore

$$I = 2 \cdot \frac{\pi}{2} = \pi.$$

$$\boxed{I = \pi.}$$

### 4.7 Problem 4(vii)

$$I = \int_0^1 \int_0^1 \sqrt{s+t} \, ds \, dt.$$

For fixed $t$,

$$\int \sqrt{s+t} \, ds = \frac{2}{3}(s+t)^{3/2}.$$

Hence

$$\int_0^1 \sqrt{s+t} \, ds = \frac{2}{3}\left((1+t)^{3/2} - t^{3/2}\right).$$

Therefore

$$I = \frac{2}{3}\int_0^1 (1+t)^{3/2} \, dt - \frac{2}{3}\int_0^1 t^{3/2} \, dt.$$

Now

$$\int_0^1 (1+t)^{3/2} \, dt = \left[\frac{2}{5}(1+t)^{5/2}\right]_0^1 = \frac{2}{5}(2^{5/2}-1),$$

and

$$\int_0^1 t^{3/2} \, dt = \frac{2}{5}.$$

Thus

$$I = \frac{2}{3}\left(\frac{2}{5}(2^{5/2}-1) - \frac{2}{5}\right) = \frac{4}{15}(2^{5/2}-2).$$

$$\boxed{I = \frac{4}{15}(2^{5/2}-2).}$$

### 4.8 Problem 4(viii)

$$I = \int_1^2 \int_0^{2z} \int_0^{\ln x} x e^y \, dy \, dx \, dz.$$

**Source/interpretation note**

The inner upper bound $\ln x$ is undefined at $x = 0$. The integral is interpreted as an improper integral at $x = 0^+$. After evaluating the inner integral, the resulting expression is integrable on $[0, 2z]$.

Inner integral:

$$\int_0^{\ln x} xe^y \, dy = x[e^y]_0^{\ln x} = x(x-1) = x^2 - x.$$

Then

$$\int_0^{2z} (x^2 - x) \, dx = \left[\frac{x^3}{3} - \frac{x^2}{2}\right]_0^{2z} = \frac{8z^3}{3} - 2z^2.$$

Finally,

$$I = \int_1^2 \left(\frac{8z^3}{3} - 2z^2\right) dz = \frac{8}{3}\left[\frac{z^4}{4}\right]_1^2 - 2\left[\frac{z^3}{3}\right]_1^2.$$

So

$$I = \frac{2}{3}(16-1) - \frac{2}{3}(8-1) = 10 - \frac{14}{3} = \frac{16}{3}.$$

$$\boxed{I = \frac{16}{3}.}$$

## 4.9 Problem 4(ix)

$$I = \int_0^{\pi/2} \int_0^y \int_0^x \cos(x+y+z) \, dz \, dx \, dy.$$

Inner integral:

$$\int_0^x \cos(x+y+z) \, dz = [\sin(x+y+z)]_0^x = \sin(2x+y) - \sin(x+y).$$

Now integrate in $x$:

$$\int_0^y \sin(2x+y) \, dx = \left[-\frac{1}{2}\cos(2x+y)\right]_0^y = -\frac{1}{2}\cos 3y + \frac{1}{2}\cos y.$$

Also,

$$\int_0^y \sin(x+y) \, dx = [-\cos(x+y)]_0^y = -\cos 2y + \cos y.$$

Therefore the integral in $x$ equals

$$-\frac{1}{2}\cos 3y + \frac{1}{2}\cos y - (-\cos 2y + \cos y) = -\frac{1}{2}\cos 3y + \cos 2y - \frac{1}{2}\cos y.$$

Finally,

$$I = \int_0^{\pi/2} \left(-\frac{1}{2}\cos 3y + \cos 2y - \frac{1}{2}\cos y\right) dy.$$

Thus

$$I = \left[-\frac{1}{6}\sin 3y + \frac{1}{2}\sin 2y - \frac{1}{2}\sin y\right]_0^{\pi/2}.$$

Evaluate at $y = \pi/2$:

$$-\frac{1}{6}\sin\frac{3\pi}{2} + \frac{1}{2}\sin\pi - \frac{1}{2}\sin\frac{\pi}{2} = -\frac{1}{6}(-1) + 0 - \frac{1}{2} = \frac{1}{6} - \frac{1}{2} = -\frac{1}{3}.$$

At $y = 0$, the value is 0. Hence

$$\boxed{I = -\frac{1}{3}.}$$

### 4.10 Problem 4(x)

$$I = \int_0^{\sqrt{\pi}} \int_0^x \int_0^{xz} x^2 \sin y \, dy \, dz \, dx.$$

Inner integral:

$$\int_0^{xz} x^2 \sin y \, dy = x^2[-\cos y]_0^{xz} = x^2(1 - \cos(xz)).$$

Then

$$\int_0^x x^2(1 - \cos(xz)) \, dz = x^2 \left[ z - \frac{\sin(xz)}{x} \right]_0^x = x^3 - x \sin(x^2).$$

Therefore

$$I = \int_0^{\sqrt{\pi}} (x^3 - x \sin(x^2)) \, dx.$$

The first integral is

$$\int_0^{\sqrt{\pi}} x^3 \, dx = \left[ \frac{x^4}{4} \right]_0^{\sqrt{\pi}} = \frac{\pi^2}{4}.$$

For the second integral, let $u = x^2$, so $du = 2x \, dx$. Then

$$\int_0^{\sqrt{\pi}} x \sin(x^2) \, dx = \frac{1}{2} \int_0^{\pi} \sin u \, du = \frac{1}{2}[-\cos u]_0^{\pi} = \frac{1}{2}(2) = 1.$$

Thus

$$\boxed{I = \frac{\pi^2}{4} - 1.}$$

## 5 Problem 5: Double integrals over specified regions

### 5.1 Problem 5(i)

Compute

$$\iint_R \sin(x - y) \, dA, \qquad R = \left\{ (x, y) : 0 \le x \le \frac{\pi}{2}, \ 0 \le y \le \frac{\pi}{2} \right\}.$$

$$I = \int_0^{\pi/2} \int_0^{\pi/2} \sin(x - y) \, dy \, dx.$$

Since

$$\frac{d}{dy} \cos(x - y) = \sin(x - y),$$

we get

$$\int_0^{\pi/2} \sin(x - y) \, dy = [\cos(x - y)]_0^{\pi/2} = \cos\left(x - \frac{\pi}{2}\right) - \cos x.$$

Because $\cos(x - \pi/2) = \sin x$,

$$I = \int_0^{\pi/2} (\sin x - \cos x) \, dx = [-\cos x - \sin x]_0^{\pi/2} = 0.$$

$$\boxed{I = 0.}$$

## 5.2 Problem 5(ii)

Compute

$$\iint_R \left(y + xy^{-2}\right) dA, \qquad R = \{(x, y) : 0 \le x \le 2,\ 1 \le y \le 2\}.$$

$$I = \int_0^2 \int_1^2 \left(y + xy^{-2}\right) dy\, dx.$$

Inner integral:

$$\int_1^2 y\, dy = \frac{3}{2}, \qquad \int_1^2 xy^{-2}\, dy = x[-y^{-1}]_1^2 = x\left(-\frac{1}{2} + 1\right) = \frac{x}{2}.$$

Thus

$$I = \int_0^2 \left(\frac{3}{2} + \frac{x}{2}\right) dx = \left[\frac{3x}{2} + \frac{x^2}{4}\right]_0^2 = 3 + 1 = 4.$$

$$\boxed{I = 4.}$$

## 5.3 Problem 5(iii)

Compute

$$\iint_R (x^2 + 2y) dA,$$

where $R$ is bounded by $y = 0$, $y = x^2$, and $x = 1$.

The region is

$$0 \le x \le 1, \qquad 0 \le y \le x^2.$$

Therefore

$$I = \int_0^1 \int_0^{x^2} (x^2 + 2y) dy\, dx.$$

Inner integral:

$$\int_0^{x^2} (x^2 + 2y) dy = \left[x^2 y + y^2\right]_0^{x^2} = x^4 + x^4 = 2x^4.$$

Thus

$$I = \int_0^1 2x^4\, dx = 2\left[\frac{x^5}{5}\right]_0^1 = \frac{2}{5}.$$

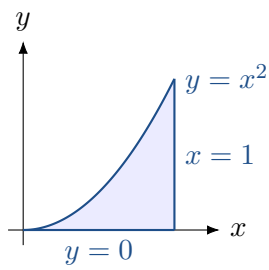



Figure 4: Region for Problem 5(iii).

$$\boxed{I = \frac{2}{5}.}$$

### 5.4 Problem 5(iv)

Compute

$$\iint_R y^2 \, dA,$$

where $R$ is the triangle with vertices $(0,1), (1,2), (4,1)$.

Use horizontal slices. The left edge from $(0,1)$ to $(1,2)$ is

$$y = x + 1 \quad \Longrightarrow \quad x = y - 1.$$

The right edge from $(1,2)$ to $(4,1)$ has slope

$$\frac{1-2}{4-1} = -\frac{1}{3}.$$

Using point $(1,2)$,

$$y - 2 = -\frac{1}{3}(x-1) \quad \Longrightarrow \quad x = 7 - 3y.$$

The vertical range is $1 \le y \le 2$. Hence

$$I = \int_1^2 \int_{y-1}^{7-3y} y^2 \, dx \, dy.$$

Inner integral:

$$\int_{y-1}^{7-3y} y^2 \, dx = y^2 \left[(7-3y) - (y-1)\right] = y^2(8-4y).$$

Therefore

$$I = \int_1^2 (8y^2 - 4y^3) dy = \left[\frac{8y^3}{3} - y^4\right]_1^2.$$

At $y = 2$, this is

$$\frac{64}{3} - 16 = \frac{16}{3}.$$

At $y = 1$, this is

$$\frac{8}{3} - 1 = \frac{5}{3}.$$

Thus

$$I = \frac{16}{3} - \frac{5}{3} = \frac{11}{3}.$$

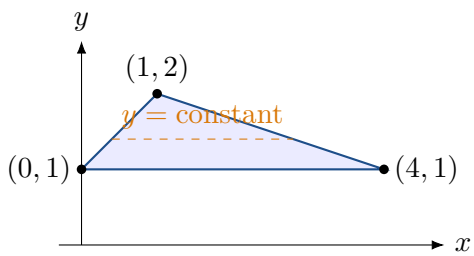



Figure 5: Horizontal slicing for Problem 5(iv).

$$\boxed{I = \frac{11}{3}.}$$

### 5.5 Problem 5(v)

Compute

$$\iint_R xy^2\,dA,$$

where $R$ is enclosed by

$$x = 0 \quad \text{and} \quad x = \sqrt{1-y^2}.$$

This is the right half of the unit disk:

$$-1 \le y \le 1, \qquad 0 \le x \le \sqrt{1-y^2}.$$

Thus

$$I = \int_{-1}^{1}\int_{0}^{\sqrt{1-y^2}} xy^2\,dx\,dy.$$

Inner integral:

$$\int_0^{\sqrt{1-y^2}} xy^2\,dx = y^2\left[\frac{x^2}{2}\right]_0^{\sqrt{1-y^2}} = \frac{1}{2}y^2(1-y^2).$$

So

$$I = \frac{1}{2}\int_{-1}^{1}(y^2-y^4)dy.$$

Since the integrand is even,

$$I = \int_0^1 (y^2-y^4)dy = \left[\frac{y^3}{3}-\frac{y^5}{5}\right]_0^1 = \frac{1}{3}-\frac{1}{5} = \frac{2}{15}.$$

$$\boxed{I = \frac{2}{15}.}$$

### 5.6 Problem 5(vi)

Compute

$$\iint_R \sin(x^2+y^2)dA,$$

where $R$ is the first-quadrant region between

$$x^2+y^2 = a^2 \quad \text{and} \quad x^2+y^2 = b^2, \qquad 0 < a < b.$$

Use polar coordinates. The region is

$$a \le r \le b, \qquad 0 \le \theta \le \frac{\pi}{2}.$$

Also,

$$x^2+y^2 = r^2, \qquad dA = r\,dr\,d\theta.$$

So

$$I = \int_0^{\pi/2}\int_a^b \sin(r^2)r\,dr\,d\theta.$$

Let $u = r^2$, so $du = 2r\,dr$. Then

$$\int_a^b \sin(r^2)r\,dr = \frac{1}{2}\int_{a^2}^{b^2}\sin u\,du = \frac{1}{2}[-\cos u]_{a^2}^{b^2} = \frac{1}{2}(\cos a^2 - \cos b^2).$$

Therefore

$$I = \frac{\pi}{2}\cdot\frac{1}{2}(\cos a^2 - \cos b^2) = \frac{\pi}{4}(\cos a^2 - \cos b^2).$$

$$\boxed{I = \frac{\pi}{4}(\cos a^2 - \cos b^2).}$$

## 5.7 Problem 5(vii)

Compute

$$\iint_R e^{-x^2-y^2}\,dA,$$

where $R$ is bounded by the semicircle

$$x = \sqrt{4-y^2}$$

and the $y$-axis.

This is the right half-disk of radius 2:

$$0 \le r \le 2, \qquad -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}.$$

Thus

$$I = \int_{-\pi/2}^{\pi/2} \int_0^2 e^{-r^2} r\,dr\,d\theta.$$

Let $u = r^2$, so $du = 2r\,dr$. Then

$$\int_0^2 e^{-r^2} r\,dr = \frac{1}{2}\int_0^4 e^{-u}\,du = \frac{1}{2}[-e^{-u}]_0^4 = \frac{1}{2}(1-e^{-4}).$$

The angular length is $\pi$, so

$$I = \pi \cdot \frac{1}{2}(1-e^{-4}) = \frac{\pi}{2}(1-e^{-4}).$$

$$\boxed{I = \frac{\pi}{2}(1-e^{-4}).}$$

## 5.8 Problem 5(viii)

Compute

$$\iint_R \arctan\frac{y}{x}\,dA,$$

where

$$R = \{(x,y) : 1 \le x^2+y^2 \le 4,\ 0 \le y \le x\}.$$

Use polar coordinates. Since

$$\arctan\frac{y}{x} = \theta$$

in the first quadrant, and the condition $0 \le y \le x$ gives

$$0 \le \theta \le \frac{\pi}{4}.$$

The annulus condition gives

$$1 \le r \le 2.$$

Thus

$$I = \int_0^{\pi/4} \int_1^2 \theta r\,dr\,d\theta.$$

Inner integral:

$$\int_1^2 \theta r\,dr = \theta\left[\frac{r^2}{2}\right]_1^2 = \frac{3}{2}\theta.$$

Therefore

$$I = \frac{3}{2}\int_0^{\pi/4} \theta\,d\theta = \frac{3}{2}\left[\frac{\theta^2}{2}\right]_0^{\pi/4} = \frac{3}{4}\cdot\frac{\pi^2}{16} = \frac{3\pi^2}{64}.$$

$$\boxed{I = \frac{3\pi^2}{64}.}$$

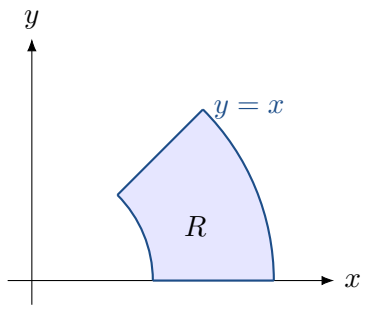



Figure 6: Polar annular sector for Problem 5(viii).

# 6 Problem 6: Volumes of solids

## 6.1 Problem 6(i)

Find the volume bounded by

$$z = 1 - x^2 - y^2$$

and the $xy$-plane.

Use polar coordinates. The $xy$-plane is $z = 0$, so the top surface must satisfy

$$1 - r^2 \geq 0 \quad \Longrightarrow \quad 0 \leq r \leq 1.$$

Thus

$$V = \int_0^{2\pi} \int_0^1 (1 - r^2) r \, dr \, d\theta.$$

Compute:

$$\int_0^1 (1 - r^2) r \, dr = \int_0^1 (r - r^3) dr = \left[ \frac{r^2}{2} - \frac{r^4}{4} \right]_0^1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}.$$

Therefore

$$V = 2\pi \cdot \frac{1}{4} = \frac{\pi}{2}.$$

$$\boxed{V = \frac{\pi}{2}.}$$

## 6.2 Problem 6(ii)

Find the volume under the cone

$$z = \sqrt{x^2 + y^2}$$

and above the disk

$$x^2 + y^2 \leq 4.$$

In polar coordinates, $z = r$ and $0 \leq r \leq 2$. Therefore

$$V = \int_0^{2\pi} \int_0^2 r \cdot r \, dr \, d\theta = \int_0^{2\pi} \int_0^2 r^2 \, dr \, d\theta.$$

So

$$V = 2\pi \left[ \frac{r^3}{3} \right]_0^2 = 2\pi \cdot \frac{8}{3} = \frac{16\pi}{3}.$$

$$\boxed{V = \frac{16\pi}{3}.}$$

### 6.3 Problem 6(iii)

Find the volume below

$$z = 18 - 2x^2 - 2y^2$$

and above the $xy$-plane.

In polar coordinates,

$$z = 18 - 2r^2.$$

The solid exists where

$$18 - 2r^2 \geq 0 \quad \Longrightarrow \quad r^2 \leq 9 \quad \Longrightarrow \quad 0 \leq r \leq 3.$$

Therefore

$$V = \int_0^{2\pi} \int_0^3 (18 - 2r^2) r \, dr \, d\theta.$$

Compute the radial integral:

$$\int_0^3 (18r - 2r^3) dr = \left[ 9r^2 - \frac{r^4}{2} \right]_0^3 = 81 - \frac{81}{2} = \frac{81}{2}.$$

Thus

$$V = 2\pi \cdot \frac{81}{2} = 81\pi.$$

$$\boxed{V = 81\pi.}$$

### 6.4 Problem 6(iv)

Find the volume enclosed by the hyperboloid

$$-x^2 - y^2 + z^2 = 1$$

and the plane

$$z = 2.$$

Rewrite the hyperboloid in cylindrical coordinates:

$$z^2 - r^2 = 1.$$

The upper sheet is

$$z = \sqrt{1 + r^2}.$$

The plane $z = 2$ intersects it when

$$2 = \sqrt{1 + r^2} \quad \Longrightarrow \quad 4 = 1 + r^2 \quad \Longrightarrow \quad r = \sqrt{3}.$$

Therefore

$$V = \int_0^{2\pi} \int_0^{\sqrt{3}} \left( 2 - \sqrt{1 + r^2} \right) r \, dr \, d\theta.$$

Radial integral:

$$\int_0^{\sqrt{3}} 2r \, dr = 3.$$

For the second part, let $u = 1 + r^2$, so $du = 2r \, dr$:

$$\int_0^{\sqrt{3}} r\sqrt{1 + r^2} \, dr = \frac{1}{2} \int_1^4 u^{1/2} \, du = \frac{1}{2} \left[ \frac{2}{3} u^{3/2} \right]_1^4 = \frac{1}{3}(8 - 1) = \frac{7}{3}.$$

Thus the radial integral is

$$3 - \frac{7}{3} = \frac{2}{3}.$$

Hence

$$V = 2\pi \cdot \frac{2}{3} = \frac{4\pi}{3}.$$

$$\boxed{V = \frac{4\pi}{3}.}$$

### 6.5 Problem 6(v)

Find the volume inside the sphere

$$x^2 + y^2 + z^2 = 16$$

and outside the cylinder

$$x^2 + y^2 = 4.$$

In cylindrical coordinates, the sphere is

$$r^2 + z^2 = 16,$$

so

$$-\sqrt{16 - r^2} \leq z \leq \sqrt{16 - r^2}.$$

Outside the cylinder $r = 2$ and inside the sphere gives

$$2 \leq r \leq 4.$$

Thus

$$V = \int_0^{2\pi} \int_2^4 \int_{-\sqrt{16-r^2}}^{\sqrt{16-r^2}} r \, dz \, dr \, d\theta.$$

Inner integral:

$$\int_{-\sqrt{16-r^2}}^{\sqrt{16-r^2}} r \, dz = 2r\sqrt{16 - r^2}.$$

Therefore

$$V = \int_0^{2\pi} \int_2^4 2r\sqrt{16 - r^2} \, dr \, d\theta.$$

Let $u = 16 - r^2$, so $du = -2r \, dr$. Then

$$\int_2^4 2r\sqrt{16 - r^2} \, dr = -\int_{12}^0 u^{1/2} \, du = \int_0^{12} u^{1/2} \, du = \left[\frac{2}{3}u^{3/2}\right]_0^{12}.$$

Now

$$12^{3/2} = 12\sqrt{12} = 24\sqrt{3}.$$

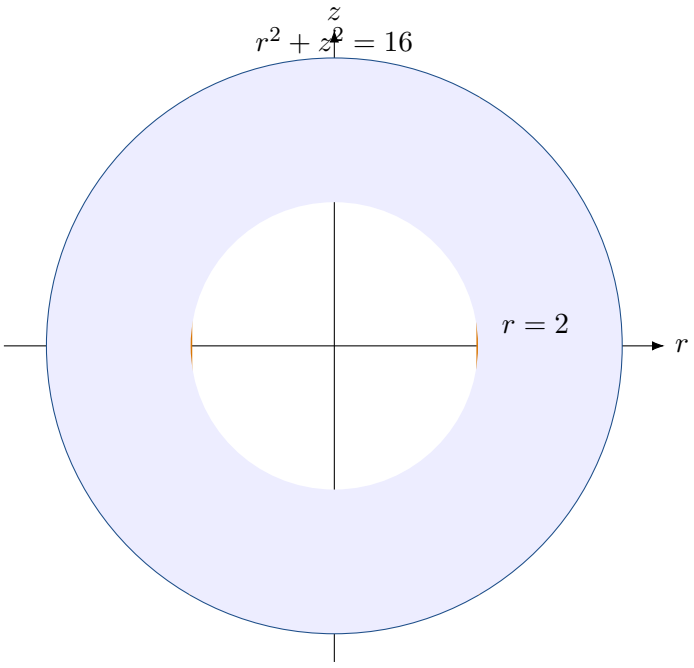
So the radial part is

$$\frac{2}{3} \cdot 24\sqrt{3} = 16\sqrt{3}.$$

Finally,

$$V = 2\pi \cdot 16\sqrt{3} = 32\pi\sqrt{3}.$$

$$\boxed{V = 32\pi\sqrt{3}.}$$




Cross-section: inside sphere and outside cylinder.

Figure 7: Meridian cross-section for Problem 6(v).

# 7 Problem 7: Triple integrals over specified solids

## 7.1 Problem 7(i)

Compute

$$\iiint_V y\,dV,$$

where

$$V = \{(x, y, z) : 0 \le x \le 3,\ 0 \le y \le x,\ x - y \le z \le x + y\}.$$

Write the integral:

$$I = \int_0^3 \int_0^x \int_{x-y}^{x+y} y\,dz\,dy\,dx.$$

Inner integral:

$$\int_{x-y}^{x+y} y\,dz = y[(x + y) - (x - y)] = 2y^2.$$

Thus

$$I = \int_0^3 \int_0^x 2y^2\,dy\,dx = \int_0^3 \left[\frac{2y^3}{3}\right]_0^x dx = \int_0^3 \frac{2x^3}{3} dx.$$

So

$$I = \frac{2}{3}\left[\frac{x^4}{4}\right]_0^3 = \frac{2}{3} \cdot \frac{81}{4} = \frac{27}{2}.$$

$$\boxed{I = \frac{27}{2}.}$$

## 7.2 Problem 7(ii)

Compute

$$\iiint_V \frac{z}{x^2 + y^2}\,dV,$$

where

$$V = \{(x, y, z) : 1 \le y \le 4,\ y \le z \le 4,\ 0 \le x \le z\}.$$

The integral is

$$I = \int_1^4 \int_y^4 \int_0^z \frac{z}{x^2 + y^2}\, dx\, dz\, dy.$$

Integrate first with respect to $x$:

$$\int_0^z \frac{z}{x^2 + y^2}\, dx = z \left[\frac{1}{y} \arctan \frac{x}{y}\right]_0^z = \frac{z}{y} \arctan \frac{z}{y}.$$

Thus an exact reduced form is

$$I = \int_1^4 \int_y^4 \frac{z}{y} \arctan \frac{z}{y}\, dz\, dy.$$

This double integral can also be reduced by changing order after integrating in $z$. Since

$$1 \le y \le 4, \qquad y \le z \le 4, \qquad 0 \le x \le z,$$

we can integrate $z$ first for fixed $(x, y)$. Then

$$z \in [\max(x, y), 4].$$

Splitting into $0 \le x \le y$ and $y \le x \le 4$, we get

$$I = \int_1^4 \left[\int_0^y \frac{16 - y^2}{2(x^2 + y^2)}\, dx + \int_y^4 \frac{16 - x^2}{2(x^2 + y^2)}\, dx\right] dy.$$

The first part is elementary:

$$\int_1^4 \int_0^y \frac{16 - y^2}{2(x^2 + y^2)}\, dx\, dy = \frac{\pi}{8} \int_1^4 \left(\frac{16}{y} - y\right) dy = \frac{\pi}{8} \left(16 \ln 4 - \frac{15}{2}\right).$$

The second part is left in a one-dimensional exact integral form:

$$\frac{1}{2} \int_1^4 \left(\frac{16}{x} - x\right) \left(\frac{\pi}{4} - \arctan \frac{1}{x}\right) dx.$$

Therefore

$$I = \frac{\pi}{8} \left(16 \ln 4 - \frac{15}{2}\right) + \frac{1}{2} \int_1^4 \left(\frac{16}{x} - x\right) \left(\frac{\pi}{4} - \arctan \frac{1}{x}\right) dx.$$

A numerical evaluation gives

$$I \approx 7.587351988.$$

**Final answer**

$$\boxed{I = \frac{\pi}{8} \left(16 \ln 4 - \frac{15}{2}\right) + \frac{1}{2} \int_1^4 \left(\frac{16}{x} - x\right) \left(\frac{\pi}{4} - \arctan \frac{1}{x}\right) dx \approx 7.587351988.}$$

### 7.3 Problem 7(iii)

Compute

$$\iiint_V \sin y\, dV,$$

where $V$ is below the plane $z = x$ and above the triangular region with vertices

$$(0, 0, 0), \quad (\pi, 0, 0), \quad (0, \pi, 0).$$

The base region in the $xy$-plane is

$$x \geq 0, \qquad y \geq 0, \qquad x + y \leq \pi.$$

The height is

$$0 \leq z \leq x.$$

Therefore

$$I = \int_0^{\pi} \int_0^{\pi - y} \int_0^{x} \sin y \, dz \, dx \, dy.$$

Inner integral:

$$\int_0^{x} \sin y \, dz = x \sin y.$$

Then

$$I = \int_0^{\pi} \int_0^{\pi - y} x \sin y \, dx \, dy = \int_0^{\pi} \sin y \left[ \frac{x^2}{2} \right]_0^{\pi - y} dy.$$

Thus

$$I = \frac{1}{2} \int_0^{\pi} (\pi - y)^2 \sin y \, dy.$$

Let $u = \pi - y$. Then $dy = -du$, and $\sin y = \sin(\pi - u) = \sin u$. The bounds change from $y = 0$ to $u = \pi$, and from $y = \pi$ to $u = 0$. Therefore

$$I = \frac{1}{2} \int_0^{\pi} u^2 \sin u \, du.$$

Integrating by parts twice gives

$$\int_0^{\pi} u^2 \sin u \, du = \pi^2 - 4.$$

Therefore

$$\boxed{I = \frac{\pi^2 - 4}{2}.}$$

## 7.4 Problem 7(iv)

Compute

$$\iiint_V xy \, dV,$$

where $V$ is bounded by the parabolic cylinders

$$y = x^2, \qquad x = y^2,$$

and the planes

$$z = 0, \qquad z = x + y.$$

The two parabolas meet at $(0, 0)$ and $(1, 1)$. In the first quadrant, the base region is

$$0 \leq x \leq 1, \qquad x^2 \leq y \leq \sqrt{x}.$$

The height is

$$0 \leq z \leq x + y.$$

Thus

$$I = \int_0^{1} \int_{x^2}^{\sqrt{x}} \int_0^{x+y} xy \, dz \, dy \, dx.$$

Inner integral:

$$\int_0^{x+y} xy \, dz = xy(x + y).$$

Therefore

$$I = \int_0^1 \int_{x^2}^{\sqrt{x}} (x^2y + xy^2)\, dy\, dx.$$

Compute the inner integral:

$$\int (x^2y + xy^2)dy = \frac{x^2y^2}{2} + \frac{xy^3}{3}.$$

Evaluate from $y = x^2$ to $y = \sqrt{x}$:

$$\left(\frac{x^2 \cdot x}{2} + \frac{x \cdot x^{3/2}}{3}\right) - \left(\frac{x^2 \cdot x^4}{2} + \frac{x \cdot x^6}{3}\right) = \frac{x^3}{2} + \frac{x^{5/2}}{3} - \frac{x^6}{2} - \frac{x^7}{3}.$$

Now integrate from 0 to 1:

$$I = \int_0^1 \left(\frac{x^3}{2} + \frac{x^{5/2}}{3} - \frac{x^6}{2} - \frac{x^7}{3}\right) dx.$$

Thus

$$I = \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{3} \cdot \frac{2}{7} - \frac{1}{2} \cdot \frac{1}{7} - \frac{1}{3} \cdot \frac{1}{8}.$$

Compute the fractions:

$$I = \frac{1}{8} + \frac{2}{21} - \frac{1}{14} - \frac{1}{24} = \frac{3}{28}.$$

$$\boxed{I = \frac{3}{28}.}$$

## 7.5 Problem 7(v)

Compute

$$\iiint_V x^2\, dV,$$

where $V$ is the tetrahedron with vertices

$$(0,0,0), \quad (1,0,0), \quad (0,1,0), \quad (0,0,1).$$

The tetrahedron is

$$x \geq 0, \qquad y \geq 0, \qquad z \geq 0, \qquad x + y + z \leq 1.$$

Thus

$$0 \leq x \leq 1, \qquad 0 \leq y \leq 1 - x, \qquad 0 \leq z \leq 1 - x - y.$$

So

$$I = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} x^2\, dz\, dy\, dx.$$

Inner integral:

$$\int_0^{1-x-y} x^2\, dz = x^2(1 - x - y).$$

Then

$$\int_0^{1-x} x^2(1 - x - y)dy = x^2 \left[(1 - x)y - \frac{y^2}{2}\right]_0^{1-x} = x^2 \cdot \frac{(1 - x)^2}{2}.$$

Therefore

$$I = \frac{1}{2} \int_0^1 x^2(1 - x)^2 dx.$$

Expand:

$$x^2(1 - x)^2 = x^2 - 2x^3 + x^4.$$

Hence

$$I = \frac{1}{2}\left(\frac{1}{3} - \frac{1}{2} + \frac{1}{5}\right) = \frac{1}{2} \cdot \frac{1}{30} = \frac{1}{60}.$$

$$\boxed{I = \frac{1}{60}.}$$

# 8 Problem 8: Change the order of integration

## 8.1 Problem 8(i)

Given

$$\int_0^1 \left[ \int_0^y f(x,y)\,dx \right] dy.$$

The region is

$$0 \le y \le 1, \qquad 0 \le x \le y.$$

This means

$$0 \le x \le y \le 1.$$

If we describe it with $x$ first, then

$$0 \le x \le 1, \qquad x \le y \le 1.$$

Therefore

$$\boxed{\int_0^1 \int_0^y f(x,y)\,dx\,dy = \int_0^1 \int_x^1 f(x,y)\,dy\,dx.}$$

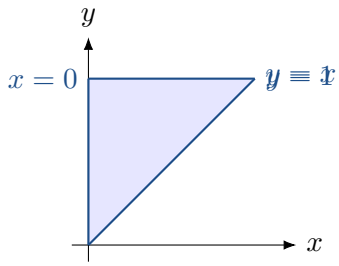



Figure 8: Region $0 \le x \le y \le 1$.

## 8.2 Problem 8(ii)

Given

$$\int_0^2 \left[ \int_{2y}^{y^2} f(x,y)\,dx \right] dy.$$

For $0 < y < 2$, we have

$$y^2 < 2y.$$

So the given integral has reversed inner bounds. The geometric region between the curves is

$$y^2 \le x \le 2y, \qquad 0 \le y \le 2.$$

The original integral is therefore the negative of the positively oriented integral over this region:

$$\int_0^2 \int_{2y}^{y^2} f(x,y)\,dx\,dy = -\int_0^2 \int_{y^2}^{2y} f(x,y)\,dx\,dy.$$

Now solve the boundary equations for $y$:

$$x = y^2 \quad \Longrightarrow \quad y = \sqrt{x},$$

and

$$x = 2y \quad \Longrightarrow \quad y = \frac{x}{2}.$$

The curves intersect at $y = 0, 2$, giving $x = 0, 4$. For $0 \le x \le 4$,

$$\frac{x}{2} \le y \le \sqrt{x}.$$

Thus

$$\boxed{\int_0^2 \int_{2y}^{y^2} f(x,y)\,dx\,dy = -\int_0^4 \int_{x/2}^{\sqrt{x}} f(x,y)\,dy\,dx.}$$

Equivalently, preserving reversed orientation inside,

$$\boxed{\int_0^2 \int_{2y}^{y^2} f(x,y)\,dx\,dy = \int_0^4 \int_{\sqrt{x}}^{x/2} f(x,y)\,dy\,dx.}$$

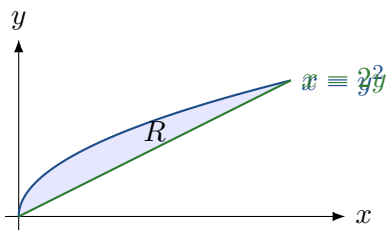



Figure 9: Region between $x = y^2$ and $x = 2y$. The original integral uses reversed inner limits.

### 8.3 Problem 8(iii)

The printed variables are inconsistent in the scan. The standard consistent reading is

$$\int_1^e \left[\int_0^{\ln x} f(x,y)\,dy\right] dx.$$

The region is

$$1 \le x \le e, \qquad 0 \le y \le \ln x.$$

The inequality $y \le \ln x$ is equivalent to

$$x \ge e^y.$$

Since $x \le e$, the maximum value of $y$ is 1. Therefore the changed order is

$$0 \le y \le 1, \qquad e^y \le x \le e.$$

Thus

$$\boxed{\int_1^e \int_0^{\ln x} f(x,y)\,dy\,dx = \int_0^1 \int_{e^y}^{e} f(x,y)\,dx\,dy.}$$

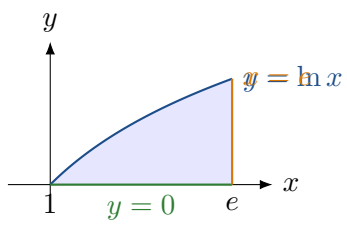



Figure 10: Region under $y = \ln x$ from $x = 1$ to $x = e$.

## 9 Problem 9: Equivalent iterated integrals

**Method / reasoning plan**

The safest way to write equivalent iterated integrals is to first translate the given integral into inequalities describing the solid. Then choose a new variable order and solve those inequalities in that order.

## 9.1 Problem 9(i)

Given

$$\int_0^1 \int_y^1 \int_0^y f(x,y,z)\,dx\,dz\,dy.$$

The inequalities are

$$0 \le y \le 1, \qquad y \le z \le 1, \qquad 0 \le x \le y.$$

Equivalently,

$$0 \le x \le y \le z \le 1.$$

Three other equivalent iterated integrals are:

$$\boxed{\int_0^1 \int_x^1 \int_y^1 f(x,y,z)\,dz\,dy\,dx.}$$

$$\boxed{\int_0^1 \int_0^z \int_0^y f(x,y,z)\,dx\,dy\,dz.}$$

$$\boxed{\int_0^1 \int_0^z \int_x^z f(x,y,z)\,dy\,dx\,dz.}$$

## 9.2 Problem 9(ii)

Given

$$\int_0^1 \int_0^{1-x^2} \int_0^{1-x} f(x,y,z)\,dy\,dz\,dx.$$

The inequalities are

$$0 \le x \le 1, \qquad 0 \le z \le 1-x^2, \qquad 0 \le y \le 1-x.$$

One immediate equivalent integral is obtained by switching the independent $y$- and $z$-integrations:

$$\boxed{\int_0^1 \int_0^{1-x} \int_0^{1-x^2} f(x,y,z)\,dz\,dy\,dx.}$$

For orders with $x$ inside, observe that for fixed $(y,z)$,

$$x \le 1-y, \qquad x \le \sqrt{1-z}.$$

Thus

$$0 \le x \le \min(1-y, \sqrt{1-z}).$$

The curve where the two upper bounds are equal is

$$1-y = \sqrt{1-z} \quad \Longleftrightarrow \quad z = 2y - y^2.$$

Therefore another equivalent expression is

$$\boxed{\int_0^1 \left[ \int_0^{2y-y^2} \int_0^{1-y} f(x,y,z)\,dx\,dz + \int_{2y-y^2}^1 \int_0^{\sqrt{1-z}} f(x,y,z)\,dx\,dz \right] dy.}$$

Using $z$ as the outer variable, the switching curve becomes

$$y = 1 - \sqrt{1-z}.$$

Thus a third equivalent expression is

$$\boxed{\int_0^1 \left[ \int_0^{1-\sqrt{1-z}} \int_0^{\sqrt{1-z}} f(x,y,z)\,dx\,dy + \int_{1-\sqrt{1-z}}^1 \int_0^{1-y} f(x,y,z)\,dx\,dy \right] dz.}$$

### 9.3 Problem 9(iii)

Given

$$\int_0^1 \int_{\sqrt{x}}^1 \int_0^{1-y} f(x,y,z)\,dz\,dy\,dx.$$

The inequalities are

$$0 \le x \le 1, \qquad \sqrt{x} \le y \le 1, \qquad 0 \le z \le 1-y.$$

The condition $\sqrt{x} \le y$ is equivalent to

$$x \le y^2.$$

Thus the solid can be written as

$$0 \le y \le 1, \qquad 0 \le x \le y^2, \qquad 0 \le z \le 1-y.$$

Three other equivalent iterated integrals are:

$$\boxed{\int_0^1 \int_0^{y^2} \int_0^{1-y} f(x,y,z)\,dz\,dx\,dy.}$$

$$\boxed{\int_0^1 \int_0^{1-z} \int_0^{y^2} f(x,y,z)\,dx\,dy\,dz.}$$

For fixed $(x,z)$, the inequalities require

$$\sqrt{x} \le y \le 1-z.$$

This is possible when

$$0 \le z \le 1-\sqrt{x}.$$

Therefore another equivalent integral is

$$\boxed{\int_0^1 \int_0^{1-\sqrt{x}} \int_{\sqrt{x}}^{1-z} f(x,y,z)\,dy\,dz\,dx.}$$

## Visual verification notes

This TeX file was written with TikZ diagrams for the main geometric regions and solid cross-sections. For verification, the compiled PDF should be rendered page-by-page. The expected visual checks are: no clipped text, no overlapping display equations, readable captions, and diagrams placed close to the computations they support.